地政總署
Lands Department

AECOM

Simplified Data Specifications for

ELEVATION AND DEPTH FSDT

Version 1.1
October 2024

Foreword

To promote better sharing and accessibility of spatial data among the community of stakeholders in Hong Kong Special Administrative Region ("HKSAR"), the Government acknowledged its commitment to establishing Common Spatial Data Infrastructure ("CSDI"). The main objective is to provide Bureaux/Departments ("B/Ds") and public and private organisations with an information infrastructure to share spatial data, support various smart city applications, and support Smart City development.

In March 2017, the Development Bureau ("DEVB") commissioned a consultancy study on the overall strategy for implementing the CSDI. With its implementation, the CSDI would benefit the Government and society in the long run, with convenient and easy access to high-quality and up-to-date spatial information. The CSDI will help to:

- Facilitate and support the development of a dynamic and flexible geospatial environment that will provide standardised interfaces for accessing and disseminating common spatial information;
- Facilitate information sharing across different B/Ds and eventually other stakeholders;
- Encourage the adoption of common standards to ensure quality, compatibility and interoperability amongst different systems; and
- Make available a wide range of spatial data and analyses to enable potential applications.

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1. Introduction

1.1. General Overview

The Simplified Data Specifications for Elevation and Depth ("the/this document") provide information about the datasets in the Elevation and Depth Framework Spatial Data Theme ("FSDT").

1.2. Glossary and Definitions

This section provides a glossary of terms and definitions, listed alphabetically, as applicable to this document.

- **Digital Surface Model**[1] ("DSM"): The three-dimensional geometry of every feature on the ground, for example, vegetation, buildings and bridges.
- **Digital Terrain Model**[1] ("DTM"): The three-dimensional shape of the Earth's surface (ground surface topography).
- **Light Detection and Ranging**[2] ("LiDAR"): A sensor that uses a laser to transmit a light pulse and a receiver with sensitive detectors to measure the backscattered or reflected light. Distance to the object is determined by recording the time between transmitted and backscattered pulses and by using the speed of light to calculate the distance travelled.

[1] D2.8.II.1 INSPIRE Data Specification on Elevation – Technical Guidelines
[2] ISO 19101-2:2018 - Geographic information – Part 2: Imagery

1.3. Acronyms and Abbreviations

This section lists the expanded description of all acronyms (words formed from the initial letters of a phrase) and abbreviations (the shortened form of a word or phrase) used in this document in alphabetical order.

Acronym/Abbreviation	Description
B/D	Bureau/Department
CEDD	Civil Engineering and Development Department
CSDI	Common Spatial Data Infrastructure
DEVB	Development Bureau
DSM	Digital Surface Model
DTM	Digital Terrain Model
FSDT	Framework Spatial Data Theme
GeoTIFF	Georeferenced Tagged Image File Format
HKSAR	Hong Kong Special Administrative Region
iB1000	Digital Topographic Map iB1000
INSPIRE	Infrastructure for Spatial Information in Europe
ISO (or ISO/TS)	International Organization for Standardization / Technical Specifications
LandsD	Lands Department
LiDAR	Light Detection and Ranging
SI	International System of Units
the/this document	Simplified Data Specifications for Elevation and Depth

2. Framework Spatial Data Definitions

2.1. Overview

Elevation and depth describe the Earth's surface on land and under a water body relative to a vertical datum. It refers to a vertical position above or below a reference surface such as sea level.

2.2. Scope & Definitions

This document covers the following sub-themes:

- **Elevation**: Elevation measures the vertical distance of a point on the Earth's surface above or below the datum.
- **Depth**: The vertical distance from a given water level to the bottom of the sea or any solid surface of the Earth covered by water.

3. Framework Spatial Data Content & Structure

3.1. Datasets under Sub-themes

There are two sub-themes within the Elevation and Depth FSDT, namely Elevation and Depth.

Elevation sub-theme covers the following dataset(s):

- Digital Terrain Model (DTM): DTM is a digital terrain model of the HKSAR. It shows the topography of terrain (including some non-ground information such as elevated roads and bridges) in 5-metre raster grid with an accuracy of ±5m.
- LiDAR Data (Raster File and Grid Index*): Acquisition of LiDAR data was commissioned by the Geotechnical Engineering Office of the CEDD covering all territories of Hong Kong including Hong Kong Island, the Kowloon Peninsula, the New Territories and Lantau Island. To this end, two sets of LiDAR data for the area were acquired from a fixed wing aircraft between 1st December 2010 and 8th January 2011 and a Helicopter between 20th December 2019 and 2nd February 2020 respectively.

*Grid Index in vector format of the corresponding LiDAR Data is uploaded to the CSDI Portals to facilitate users in accessing the download link of the raster file.

Depth sub-theme does not contain any dataset(s).

3.2. Physical Data Model

This section describes the physical data model for each sub-theme within the Elevation and Depth FSDT.

The data model is not applicable for the raster datasets, namely the Digital Terrain Model (DTM) and LiDAR Data (Raster File).

However, the data model of the corresponding grid index of the raster datasets, LiDAR Data (Grid Index), is shown below.

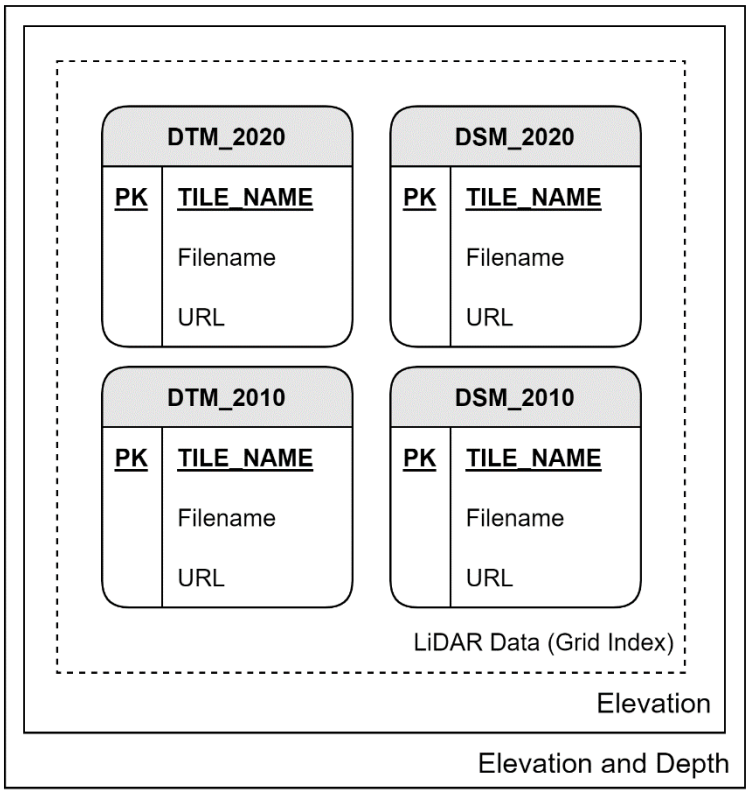

Figure 3-1 Physical Data Model of Elevation and Depth FSDT

3.3. Entity Description

This section provides a detailed description of each physical entity illustrated in the physical data model in Section 3.2.

3.3.1. Elevation – LiDAR Data (Grid Index)

3.3.1.1. DSM_2010

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
TILE_NAME	String, 50	Tile name	Y
Filename	String, 100	File name	Y
URL	String, 200	Download link of GeoTIFF of the corresponding tile	Y

3.3.1.2. DTM_2010

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
TILE_NAME	String, 50	Tile name	Y
Filename	String, 100	File name	Y
URL	String, 200	Download link of GeoTIFF of the corresponding tile	Y

3.3.1.3. DSM_2020

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
TILE_NAME	String, 7	Tile name	Y
Filename	String, 100	File name	Y
URL	String, 200	Download link of GeoTIFF of the corresponding tile	Y

3.3.1.4. DTM_2020

Entity type: Feature Layer

Field Name (Alias Name, optional)	Data Type	Description	Allow Null Value (Y/N)
TILE_NAME	String, 7	Tile name	Y
Filename	String, 100	File name	Y
URL	String, 200	Download link of GeoTIFF of the corresponding tile	Y

3.4. Unique Identifiers

The following table describes the unique identifier of each entity in the Elevation and Depth FSDT.

Data Layer	Unique Identifier	Description
DSM_2010	TILE_NAME	A unique identifier maintained by CEDD based on their internal practice, consists of four segments, for example, "e801n807". The following describes the content of each segment: • "e": One letter denoting Easting. • "801": First three digits of the Easting of the lower left corner of the grid. • "n": One letter denoting Northing. • "807": First three digits of the Northing of the lower left corner of the grid.
DTM_2010		
DSM_2020	TILE_NAME	A unique identifier maintained by CEDD which is based on the iB1000 Map Index, consists of four segments, for example, "11NE14D". The following describes the content of each segment: • "11": One to two digit(s), no padding zero. • "NE": Two letters denoting the four quadrants that the grid lies in, for example, NW for Northwest, NE for Northeast, SW for Southwest and SE for Southeast. • "14": One to two digit(s), no padding zero. • "D": One letter between A-D, denoting the subdivided grid number.
DTM_2020		

3.5. Geographical Coverage

The dataset(s) specified in this document covers the entire territories of HKSAR.

The tiling scheme of the data specified in this document is/are:

- Digital Surface Model from 2010 LiDAR Survey (Raster File and Grid Index): The dataset is tiled into 1,574 tiles. The tile size is 1,000 m * 1,000 m.
- Digital Terrain Model from 2010 LiDAR Survey (Raster File and Grid Index): The dataset is tiled into 1,574 tiles. The tile size is 1,000 m * 1,000 m.
- Digital Surface Model from 2020 LiDAR Survey (Raster File and Grid Index): The dataset is tiled into 3,315 tiles. The tile size is 750 m (W) * 600 m (H). The tiling scheme follows the iB1000 tiling scheme.
- Digital Terrain Model from 2020 LiDAR Survey (Raster File and Grid Index): The dataset is tiled into 3,253 tiles. The tile size is 750 m (W) * 600 m (H). The tiling scheme follows the iB1000 tiling scheme.

3.6. Units of Measurement

The units of measurement used in all datasets under Elevation and Depth FSDT refer to the International System of Units ("SI").

3.7. Spatial Resolution/Scale

The spatial resolution(s) or scale(s) of the dataset(s) specified in this document is/are:

- Digital Terrain Model (DTM): The dataset is stored in raster form at 5 meters of grid size.
- Digital Surface Model and Digital Terrain Model from 2010 and 2020 LiDAR Survey (Raster File): The dataset is stored in raster form at 0.5 meters of grid size.
- Digital Surface Model and Digital Terrain Model from 2010 LiDAR Survey (Grid Index): The dataset is stored in vector form at a spatial resolution of 1,000 m x 1,000 m.
- Digital Surface Model and Digital Terrain Model from 2020 LiDAR Survey (Grid Index): The dataset is stored in vector form at a spatial resolution of 750 m x 600 m.

3.8. Spatial Representation Type

The spatial representation type(s) of the dataset(s) specified in this document is/are:

- All datasets are represented as raster images to represent the geographical location of the entity.
- All the grid indexes of the LiDAR Data are represented as polygons with standard dimensions to represent the geographical location of the entity.

3.9. Spatial Reference System

The spatial data covered in this document are geo-referenced or projected to one of the Coordinate Reference Systems as specified in the Simplified Data Specifications for Coordinate Reference System of the CSDI.

3.10. Relationships and Dependencies with Other FSDTs

There are no relationships between Elevation and Depth FSDT and other FSDTs.

The English version shall prevail in case of any discrepancy or inconsistency between the English version and its Chinese translation.